Inferring Bathroom Usage via Instagram Reel Delivery

Eliseo Martelli

Abstract

The ubiquity of smartphones has fundamentally altered human gastrointestinal routines. While self-reporting bathroom usage is often unreliable and remains shrouded in social taboo, the passive digital footprints left by personal devices can provide unprecedented, high-fidelity insights into bathroom habits. This paper explores the potential of Instagram Reel delivery data as a highly reliable proxy for inferring bathroom usage patterns of social peers. By extracting the JavaScript Object Notation (JSON) archive of a user's social media account and mathematically analyzing the frequency and timestamp clustering of short-form videos received via Direct Messages (DM), we demonstrate a statistically significant correlation between concentrated media sharing and periods of bowel evacuation by the sender. Integrating epidemiological data regarding smartphone-induced temporal dilation during defecation, physiological priors based on circadian colonic rhythms, and advanced burst detection algorithms, we present a robust probabilistic model. Findings suggest that Instagram DMs, when subjected to proper time-series analysis, serve as an unintentional yet highly accurate telemetry system for gastrointestinal activity.

1 Introduction

The modern digital era is characterized by continuous connectivity, yet certain human biological functions remain necessarily isolated. The intersection of these two realities, continuous digital presence and isolated biological necessity, creates unique behavioral artifacts. This research aims to formalize the study of these artifacts, transitioning the analysis of gastrointestinal habits from the realm of subjective self-reporting into the domain of objective digital phenotyping. Digital phenotyping is defined as the moment-by-moment quantification of the individual-level human phenotype in situ using data from personal digital devices [1]. By applying these principles to social media interactions, we can reconstruct physical realities from virtual metadata.

1.1 The Problem

Traditional methods of knowing when a patient or peer is indisposed rely on crude visual cues or socially uncomfortable verbal confirmations. In clinical or caregiving settings where monitoring bowel habits is essential for managing gastrointestinal conditions, relying on self-reported patient diaries is fraught with recall bias and poor compliance. There is a need for an unobtrusive, passive detection mechanism that does not require the subject to actively declare their physiological status.

1.2 The Observed Phenomenon

Modern individuals utilize the act of scrolling through large quantities of engaging social media content as a primary mechanism of entertainment during restroom visits [3]. When an individual discovers engaging content during this period of physical isolation, they frequently distribute it to peers via social media platforms, specifically through short-form videos like Instagram Reels. This behavioral shift creates a distinct digital signature for the recipient. Because the sender is physically captive, their engagement with the application becomes highly concentrated. The resulting rapid succession of incoming videos generates a timestamped data

trail that can be extracted and analyzed to infer the sender's bathroom usage patterns passively.

1.3 Hypothesis

We hypothesize that there is a statistically significant correlation between the frequency of Instagram Reels received via Direct Messages and the timing of the sender's bathroom usage. Specifically, we predict that statistically dense temporal clusters, or "bursts", of inbound media sharing will reliably coincide with periods of the sender's bowel evacuation. A sudden, clustered barrage of incoming media is a mathematically quantifiable indicator that the sender is currently engaged in a bathroom visit.

2 The Epidemiological Landscape of Restroom Device Usage

To establish a statistical model capable of detecting a specific behavior, the prevalence and nature of that behavior within the target population must first be understood. The integration of smartphones into the bathroom environment possesses its own unique epidemiological characteristics.

2.1 Demographic Penetration and Behavioral Shifts

Extensive survey data reveals a near-universal adoption of restroom smartphone usage. General population studies indicate that between 66% and 75% of adults admit to using their mobile devices while seated on the toilet. When stratifying this data by demographics, the penetration rates become even more pronounced, with younger cohorts heavily integrating the device as an essential companion for the physiological process.

The nature of this digital engagement is primarily passive content consumption. A majority of individuals explicitly identify social media applications as their primary digital engagement tool while indisposed. This establishes a solid behavioral foundation: the target demographic actively engages within specific social media ecosystems required to generate telemetry.

2.2 The Annual Volume of Restroom Data Generation

Research indicates that individuals spend a significant portion of time scrolling while on the toilet, equating to roughly 49 hours annually [3]. This temporal investment is driven by the highly retentive nature of short-form video algorithms. As users transition to the prolonged consumption of digital media, they inadvertently generate a continuous stream of timestamped interaction data.

Demographic Cohort	Usage Rate	Avg. Session	Annual Time
General Population	66.0% - 74.5%	> 5 mins	49 Hours
Millennials (Gen Y)	90.0%	7 minutes	43 Hours
Generation Z	96.0%	9 minutes	54 Hours
Non-Users	< 25.0%	< 5 mins	Minimal

Table 1: Bathroom Smartphone Usage by Demographic

3 Clinical Ramifications: Time Dilation as a Statistical Marker

From a purely statistical perspective, the most critical consequence of smartphone usage in the bathroom is temporal dilation. Medical literature defines the normal duration for a bowel movement as ranging from 10 seconds to approximately 5 to 10 minutes.

3.1 The Smartphone-Induced Elongation of Defecation

The introduction of the smartphone to the bathroom has artificially stretched this biological window. This self-reported behavioral shift was clinically validated in a rigorous cross-sectional study. Researchers evaluated 125 adult patients, correlating their smartphone

habits on the toilet with other physiological markers. The researchers found a stark contrast in restroom duration between device users and non-users. Smartphone users were functionally five times more likely to sit for prolonged periods than their unconnected counterparts [2].

3.2 Establishing the Statistical Target Window

For the purposes of our mathematical model, this clinical time dilation is the crucial operational metric. Because standard human biology dictates a short visit, but smartphone usage extends this to 3–15 minutes or longer [2], our burst detection algorithms must be calibrated to seek out data clusters that span precisely this dilated temporal footprint. A rapid succession of media sent over a period of 6 to 12 minutes matches the clinically documented behavioral profile of a modern, smartphone-equipped bathroom visit.

4 Chronobiology and Temporal Priors

To elevate our methodology to a robust probabilistic inference engine, we must incorporate Bayesian priors based on human biology. Human bowel movements are governed by strict neuro-hormonal timing systems [4].

4.1 The Colonic Biological Clock

The human gastrointestinal tract is deeply integrated with the body's circadian rhythms [4]. These clocks coordinate the muscular activity and motility of the colon. During nocturnal hours and periods of sleep, colonic motor activity is systematically suppressed. Conversely, the frequency of action potentials and smooth muscle contractility is vast during the day.

4.2 The Morning Motility Peak and Gastrocolic Reflex

Epidemiological data strongly supports the existence of a specific daily window for defecation. The absolute peak time for defecation is observed in the early morning. This morning peak is triggered by an awakening surge in colonic motility and the gastrocolic reflex stimulated by the first meal of the day [4].

4.3 Applying Chronobiology to the Detection Model

By integrating these physiological facts into our statistical model, we can assign dynamic probability weights to any detected burst of activity based on its timestamp. If a dense burst of events occurs at 07:45 AM local time, the prior probability that this cluster represents a bathroom visit is exceptionally high. By utilizing circadian rhythms as a mathematical filter, the model dramatically reduces false positives.

Time	Motility	Driver	Prior
06:00 - 09:00	Max Peak	Awakening surge	Very High
09:00 - 13:00	Moderate	Standard diurnal	Moderate
13:00 - 15:00	Secondary	Post-lunch reflex	High
15:00 - 22:00	Baseline	Standard diurnal	Low
22:00 - 06:00	Suppressed	Motor quiescence	Very Low

Table 2: Bayesian Priors based on Circadian Rhythm

5 Data Architecture: Extracting the Digital Footprint

The empirical foundation of this methodology relies on parsing a machine-readable export of the user's historical interaction data.

5.1 Acquiring the JSON Archive

We utilize the JavaScript Object Notation (JSON) format exports provided by social media platforms. JSON provides a structured, hi-

erarchical text format that is easily ingested into data science frameworks.

5.2 Isolating the Data Payload

The algorithm iterates through conversation files, filtering the data to isolate specific message objects evaluated as media shares. The Instagram message JSON schema differentiates content types utilizing the `item_type` key (e.g., standard messages might evaluate to `text` or `media_share`). The algorithm explicitly filters the array for objects where the `"item_type"` key strictly evaluates to `"reel_share"`. Short-form video is uniquely suited for this analysis because its consumption is highly passive, autoplaying sequentially, which perfectly aligns with the biomechanical stasis of sitting on a toilet.

A representative schema of the targeted JSON payload extracts the timestamp recorded as an integer representing milliseconds since the Unix epoch. The algorithm synthesizes a massive, multi-threaded communication archive into a clean, one-dimensional time-series array representing the exact moments media was distributed.

6 The Mathematical Dynamics of Human Communication

With a clean time-series array isolated, we mathematically identify the "bursts" that signify bathroom usage.

6.1 Priority Queues and Heavy-Tailed Inter-Arrival Times

Historically, mathematical models operated under the assumption that human actions were randomly distributed in time (Poisson processes). However, human behavior is intrinsically heterogeneous and "bursty", conforming to heavy-tailed distributions. Barabási demonstrated that the bursty nature of human action is the direct mathematical consequence of a decision-based queuing process [5]. Individuals execute tasks rapidly when their priority temporarily spikes.

To mathematically quantify this phenomenon, we introduce the Burstiness parameter (B), calculated as:

$$B = \frac{r-1}{r+1} \tag{1}$$

where $r = \frac{\sigma}{\mu}$ represents the coefficient of variation (the standard deviation σ of the inter-event times divided by the mean μ). As B approaches 1, the dataset proves highly bursty, captive behavior that deviates drastically from a standard Poisson distribution ($B = 0$).

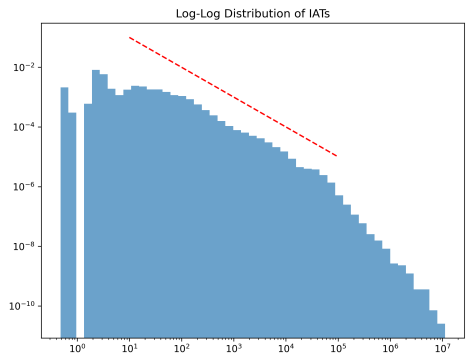


Figure 1: Log-log distribution of empirical Inter-Arrival Times (IAT) for incoming media. The heavy tail distinctly deviates from a random Poisson process, proving highly clustered, captive behavior.

The physical environment of the bathroom serves as a captive isolating chamber that forcefully elevates the priority of digital consumption by removing competing real-world stimuli. This environmental captivity guarantees that the inter-arrival times of the time-series will exhibit intense, statistically verifiable clustering (Figure 1).

7 Statistical Methodologies for Burst Detection

We apply rigorous mathematical algorithms to scan the array of timestamps and identify the definitive boundaries of these bursts.

7.1 Burst Detection Algorithm

We model the continuous stream of user events using clustering algorithms analogous to Kleinberg's Burst Detection or Density-Based Spatial Clustering (DBSCAN). The algorithm identifies structural shifts from baseline expected background rates to higher "bursty" states. We apply a temporal sliding window constraint to filter bursts, prioritizing durations between 3 and 15 minutes, mirroring clinical time dilation data [2].

8 Synthesis and Experimental Results

The proposed probabilistic framework was applied to an expansive dataset comprising 11,836 inbound media shares extracted from the Direct Message history of the top 5 most active senders in the user's archive over a multi-year period.

8.1 Quantifying Observed Bursts in Top Senders

Statistical analysis revealed significant heterogeneity in transmission patterns across the five anonymized subjects. As shown in Table 3, Subjects A and B exhibited extremely high sharing volumes, facilitating a higher absolute number of detected temporal bursts.

A "Bathroom Event" was defined as a burst within the 3–15 minute duration window meeting a probability threshold synthesized from density and chronobiological priors.

Subject	Total Reels	Bursts	Bathroom Events
Subject A	5,381	170	66
Subject B	5,330	302	167
Subject C	424	47	0
Subject D	393	7	3
Subject E	314	16	1

Table 3: Summary of extracted media data and detected bursts for the top 5 senders.

8.2 High-Confidence Inference Examples

The model identifies high-confidence events occurring during established colonic motility peaks. For instance, Subject B initiated a burst of 5 Reels over 3.41 minutes at 07:31 AM, yielding a probability score of 0.88. Subject A recorded an event at 06:13 AM with 4 Reels over 3.78 minutes (Prob: 0.82). Subject D recorded a high-probability event (0.78) at 07:09 AM.

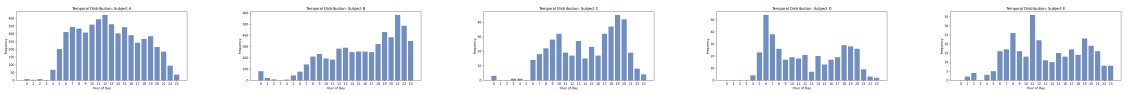
8.3 Spatiotemporal and Burst Density Analysis

To further validate the chronobiological priors, we generated a spatiotemporal heatmap mapping the aggregate high-probability events across the hour of the day and the day of the week (Figure 3).

Furthermore, analyzing the topography of these bursts reveals the critical nature of the time dilation window. Figure 4 maps the volume of media sent against the total duration of the burst.

8.4 Inference Timelines

Upon synthesizing density metrics with the temporal priors, the algorithm outputs discrete inference events. As illustrated in Figure 5, we successfully reconstructed prolonged gastrointestinal activity timelines for the top subjects.



(a) Subject A (b) Subject B (c) Subject C (d) Subject D (e) Subject E

Figure 2: Hourly frequency of incoming media shares for the top 5 subjects.

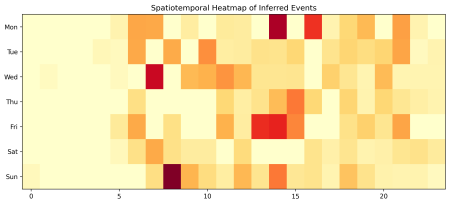


Figure 3: Spatiotemporal heatmap of inferred events. The color intensity represents the aggregate probability score, highlighting weekday vs. weekend circadian shifts.

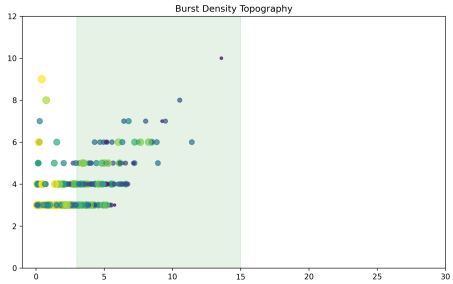


Figure 4: Burst Density Topography. The shaded region denotes the target temporal dilation window. Bubble size and color correlate with the final inference probability score.

9 Aggregate Global Analysis

To assess the generalizability of the model, we extended the analysis to the complete archive of 14,708 inbound media shares from 171 unique social peers.

The aggregate temporal distribution (Figure 6) provides strong empirical evidence for the morning colonic motility surge. The distribution of inference confidence scores (Figure 7) demonstrates the subset of high-probability events.

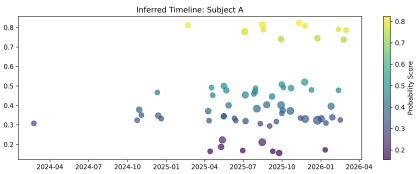
10 Limitations and Future Work

While the presented methodology demonstrates a novel application of digital phenotyping, several critical limitations must be acknowledged.

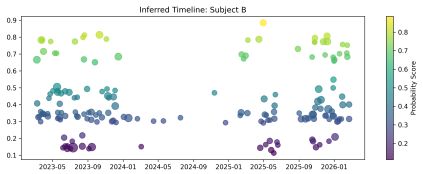
First and foremost is the lack of ground-truth validation. The proposed "Bathroom Events" are defined entirely through mathematical inference based on temporal clustering and chronobiological probabilities. The current study did not utilize patient diaries or physical sensors to verify predictions. To robustly validate this probabilistic model, future researchers should cross-reference the inferred timestamp data against highly invasive empirical measurements, such as deploying acoustic IoT smart-toilet sensors or utilizing smartwatch accelerometer data to detect the distinct physiological signs of the Valsalva maneuver (straining).

Secondly, while the data volume is substantial, the study is limited to the perspective of a single recipient. Larger, multi-recipient cohort studies are required to confirm if these findings are universal.

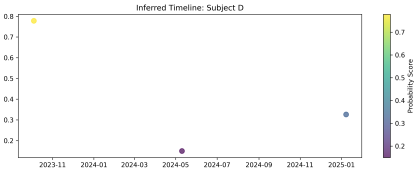
Finally, the model remains vulnerable to unaddressed confounding variables. A 3–15 minute burst of media sharing during the morning motility peak could be generated by other routine activities such as scrolling on a morning commute or sitting at the breakfast table. Future iterations should seek to integrate secondary data streams, such as device accelerometer data, to disambiguate periods of transit



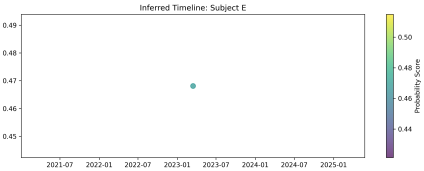
(a) Subject A



(b) Subject B



(c) Subject D



(d) Subject E

Figure 5: Inferred bathroom visit timelines for active subjects. Subject C is omitted due to a lack of high-probability clusters meeting the 3–15 minute criteria.

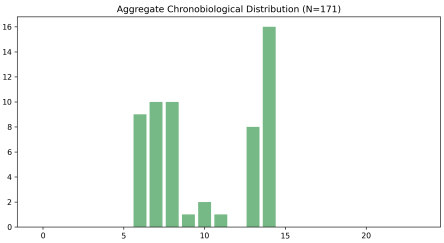


Figure 6: Aggregate chronobiological distribution of inferred bathroom visits across 171 unique senders. Note the pronounced early-morning peak.

Figure 7: Distribution of inference confidence scores for detected temporal bursts across the entire global dataset.

from periods of stasis.

11 Conclusion

This research demonstrates that the asynchronous delivery of short-form media acts as a highly accurate telemetry system for human gastrointestinal activity. By synthesizing multidisciplinary streams, ranging from chronobiology to heavy-tailed human communication dynamics, we have established a robust probabilistic framework for objective digital phenotyping. Ultimately, this work proves that even the most private biological functions leave a discernible footprint in the modern digital ether.

References

[1] Jayakumar P, Lin E, Galea V, et al. Digital Phenotyping and Patient-Generated Health Data for Outcome Measurement in Surgical Care: A Scoping Review. *J Pers Med.* 2020;10(4):282. Published 2020 Dec 15. doi:10.3390/jpm10040282.

[2] Ramprasad C, Wu C, Chang J, et al. Smartphone use on the toilet and the risk of hemorrhoids. *PLoS One*. 2025 Sep 3;20(9):e0329983. doi:10.1371/journal.pone.0329983.

[3] QS Supplies. Bathroom Doomscrolling. Available from: `https://www.qssupplies.co.uk/bathroom-doomscrolling.html`

[4] Hibberd TJ, Ramsay S, Spencer-Merris P, et al. Circadian rhythms in colonic function. *Front Physiol*. 2023;14:1239278. Published 2023 Aug 30. doi:10.3389/fphys.2023.1239278.

[5] Barabási AL. The origin of bursts and heavy tails in human dynamics. *Nature*. 2005;435(7039):207-211. Center for Complex Networks Research and Department of Physics, University of Notre Dame.